

DeFi Super-App

Full-Stack Decentralized Protocol

Deployed on Base Sepolia L2 Testnet

Why DeFi Super-App?

The Problem

-  Fragmented DeFi ecosystem requiring multiple protocols
-  Complex user experience across different platforms
-  Lack of integrated governance and security standards

Our Solution

-  Unified protocol combining AMM, Lending, and Yield Vault
 -  Single governance layer with 2-day Timelock protection
 -  Production-grade security with Chainlink oracles
- Fully indexed via The Graph for transparency

System Architecture

Core Components

- AMM (Automated Market Maker) + LP Token
- Lending Pool with collateralization
- ERC-4626 Yield Vault
- Chainlink Price Oracles

Governance Layer

- Governor + TimelockController
- ERC20Votes token-based voting

Infrastructure

- The Graph indexing (7 entities, 6 queries)
- UUPS upgradeable Treasury (V1 → V2)

Automated Market Maker

Functionality

-  Constant product formula ($x * y = k$)
-  Liquidity provision with LP tokens
-  Swap execution with slippage protection

Key Metrics

-  ReentrancyGuard on all state-changing functions
-  SafeERC20 for token transfers
-  CEI pattern enforced

Contract

0xeD03483511cd41Ba895BE5c04B3AF1a215B58D23

 Verified on Basescan

Lending Protocol & Yield Vault

Lending Pool



Collateralized lending



Interest rate model

ERC-4626 Yield Vault



Standardized tokenized vault



Automated yield optimization



Share-based accounting

```
contract YieldVault is ERC4626 {
    using SafeERC20 for IERC20;

    constructor(IERC20 _token) ERC4626(_token) {}

    function deposit(uint256 assets, address receiver) public returns (uint256) {
        require(assets > 0, "Zero assets");
        receiver = msg.sender;
    }
}
```

Built with Modern Tools

Frontend

HTML/CSS/JS

ethers v6

MetaMask

Indexing

The Graph subgraph (7 entities, 6 documented queries)

Upgradability

UUPS proxy pattern for Treasury

Oracles & Pricing

Chainlink price feeds with staleness guard

Smart Contracts

Foundry framework

Solidity 0.8.24

OpenZeppelin 5.x libraries

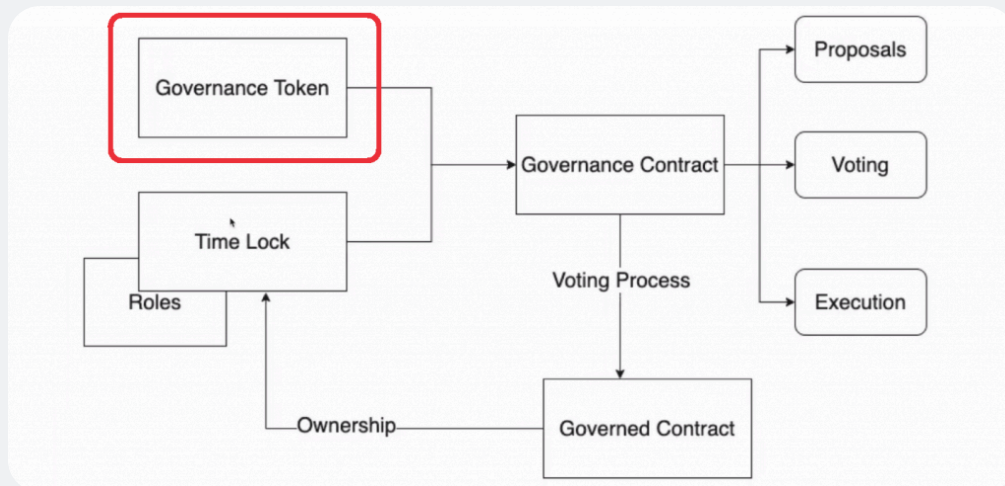
Deployment

Base Sepolia L2 testnet

CI/CD

GitHub Actions: build, test, coverage $\geq 90\%$, slither, solhint, prettier

Decentralized Governance



Governance Parameters

-  1-day voting delay
-  1-week voting period
-  4% quorum requirement
-  1% proposal threshold
-  2-day Timelock execution delay

Security Measures

-  No EOA has admin rights post-deployment
-  All admin roles held by Timelock
-  OpenZeppelin AccessControl on all privileged functions
-  Chainlink oracles revert on staleness
-  ReentrancyGuard + CEI pattern enforced
-  SafeERC20 for all token operations